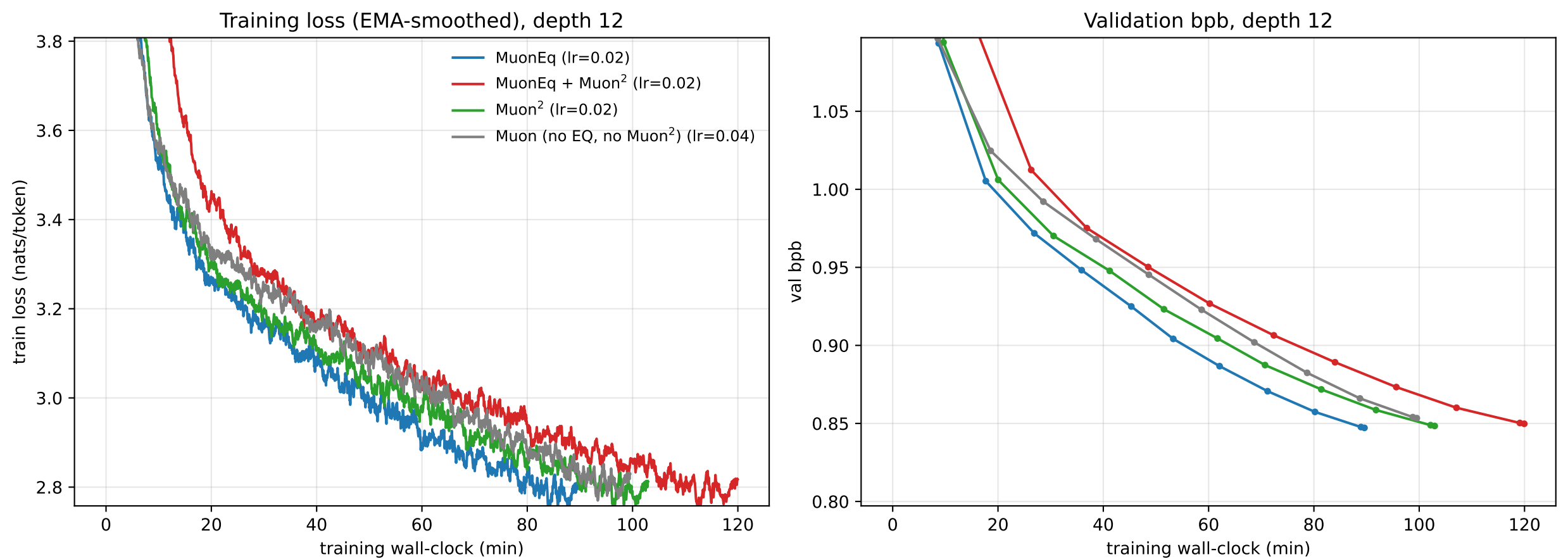

Training loss (EMA-smoothed), depth 12
train loss (nats/token)
training wall-clock (min)
MuonEq (lr=0.02)
MuonEq + Muon² (lr=0.02)
Muon² (lr=0.02)
Muon (no EQ, no Muon²) (lr=0.04)
Validation bpb, depth 12
val bpb
training wall-clock (min)